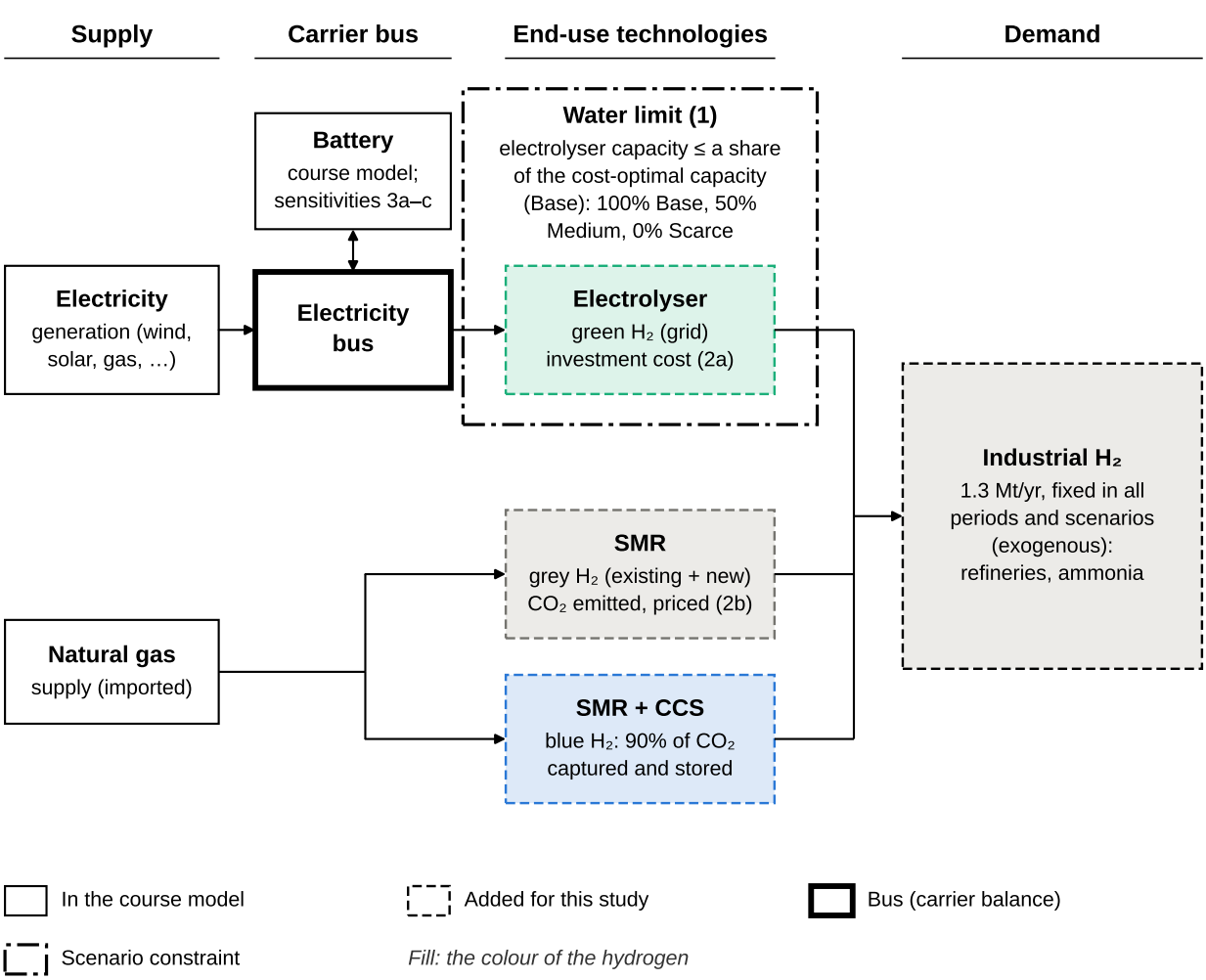
Supply
Carrier bus
End-use technologies
Demand
Battery
course model;
sensitivities 3a–c
Water limit (1)
electrolyser capacity ≤ a share
of the cost-optimal capacity
(Base): 100% Base, 50%
Medium, 0% Scarce
Electricity
generation (wind,
solar, gas, …)
Electricity
bus
Electrolyser
green H2 (grid)
investment cost (2a)
Industrial H2
1.3 Mt/yr, fixed in all
periods and scenarios
(exogenous):
refineries, ammonia
SMR
grey H2 (existing + new)
CO2 emitted, priced (2b)
Natural gas
supply (imported)
SMR + CCS
blue H2: 90% of CO2
captured and stored
In the course model
Added for this study
Bus (carrier balance)
Scenario constraint
Fill: the colour of the hydrogen